

CFB VALCARTIER, QC, Canada

WMO: 717160

Lat: 46.9004N

Lon: 71.5032W

Elev: 168

StdP: 99.33

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.6	(c) -27.2	(d) -35.3	(e) 0.1	(f) -29.9	(g) -32.4	(h) 0.2	(i) -26.8	(j) 9.1	(k) -5.4	(l) 8.1	(m) -7.5	(n) 0.8	(o) 230	(p) 0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	28.5	20.3	26.9	19.5	25.3	18.6	22.1	26.1	21.0	24.6	20.0	23.6	4.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.8	24.1	19.8	14.8	23.0	18.7	13.8	22.1	65.8	26.1	61.7	24.7	58.2	23.6	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.1	7.1	6.2	-36.0	31.2	3.7	1.4	-38.7	32.2	-40.8	33.0	-42.9	33.8	-45.5	34.8
			-36.0	23.8	3.6	1.4	-38.6	24.8	-40.8	25.6	-42.8	26.4	-45.4	27.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-13.1	-11.6	-5.6	2.6	10.6	15.7	18.5	17.4	12.8	6.2	-0.9	-8.5
	DBStd	12.17	7.76	7.11	6.49	4.27	4.06	3.62	2.76	2.99	3.92	3.99	5.11	6.67
	HDD10.0	3119	716	603	483	225	42	2	0	0	17	129	328	574
	HDD18.3	5404	974	837	741	473	242	94	31	54	172	376	578	833
	CDD10.0	841	0	0	0	2	60	174	264	229	99	12	1	0
	CDD18.3	84	0	0	0	0	2	16	37	24	4	0	0	0
	CDH23.3	847	0	0	0	1	56	208	326	207	49	0	0	0
	CDH26.7	154	0	0	0	0	10	45	61	31	8	0	0	0
(14) Wind	WSAvg	2.4	2.4	2.5	2.7	2.8	2.6	2.4	2.2	2.0	1.9	2.3	2.3	2.2
(15) Precipitation	PrecAvg	1223	87	71	73	85	95	113	141	117	123	116	96	103
	PrecMax	1406	141	117	133	194	154	209	239	176	232	234	158	226
	PrecMin	989	39	27	20	19	51	50	70	37	62	40	35	46
	PrecStd	123	30	22	26	43	29	43	43	30	42	44	28	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	5.0	10.5	19.7	28.1	30.1	30.7	29.2	27.8	21.1	13.7	5.4
		MCWB	3.7	2.6	5.3	10.7	17.1	21.5	22.5	21.1	20.7	15.5	11.9	4.8
	2%	DB	2.3	2.8	6.9	15.5	24.8	27.9	28.3	27.4	24.6	17.6	10.3	3.3
		MCWB	1.5	0.7	2.9	8.4	15.3	19.6	20.7	20.2	18.7	13.4	8.4	2.3
	5%	DB	0.6	1.2	4.7	12.4	22.0	25.8	26.6	25.7	22.2	15.0	7.9	1.8
		MCWB	-0.3	-0.1	1.8	6.7	14.0	18.4	19.8	19.1	17.4	11.7	6.0	0.9
	10%	DB	-1.7	-0.3	3.0	9.9	19.2	23.6	25.1	24.0	20.1	12.8	6.0	0.6
		MCWB	-2.7	-1.5	0.9	5.4	12.6	17.0	18.9	18.1	16.5	10.3	4.6	-0.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	3.6	6.6	13.4	19.7	23.3	23.9	22.7	21.7	16.8	12.6	4.9
		MCDB	4.3	4.4	8.5	18.6	24.9	28.2	28.4	27.2	25.3	19.6	13.4	5.2
	2%	WB	1.6	1.1	3.8	9.3	17.2	21.1	22.4	21.4	19.9	14.4	9.0	2.6
		MCDB	2.1	2.2	5.7	13.3	22.0	25.3	26.3	25.0	22.9	16.7	9.9	3.1
	5%	WB	-0.2	0.0	2.4	7.6	15.1	19.7	21.2	20.5	18.5	12.6	6.4	1.1
		MCDB	0.6	1.1	4.2	11.6	19.3	23.8	24.8	23.8	21.2	14.5	7.4	1.5
	10%	WB	-2.6	-1.7	1.1	5.8	13.5	18.2	20.1	19.4	16.9	10.7	4.7	0.0
		MCDB	-1.7	-0.4	2.7	9.0	18.4	22.3	23.7	22.9	19.5	12.3	6.0	0.6
(35) Mean Daily Temperature Range	MDBR		11.5	12.8	11.6	10.6	12.7	12.2	11.2	11.6	11.6	9.6	7.9	9.4
	5% DB	MCDBR	10.1	10.0	12.2	16.0	17.6	15.9	13.8	13.8	14.3	13.9	10.7	7.1
		MCWBR	9.8	9.0	9.1	10.4	9.8	8.6	7.3	7.5	8.9	9.9	9.2	6.9
	5% WB	MCDBR	9.8	9.2	10.2	13.7	14.7	13.3	11.7	11.5	12.3	12.0	9.1	7.0
		MCWBR	9.8	8.4	8.1	9.7	9.0	8.1	6.9	7.0	8.5	9.4	8.6	7.1
(40) Clear-Sky Solar Irradiance	taub		0.263	0.280	0.320	0.346	0.362	0.396	0.406	0.389	0.365	0.335	0.304	0.285
	taud		2.306	2.248	2.218	2.356	2.409	2.308	2.297	2.347	2.409	2.487	2.477	2.297
	Ebn at Noon		833	890	899	909	903	869	854	855	845	822	783	750
	Edh at Noon		74	98	119	114	113	126	125	115	99	78	65	68
(44) All-Sky Solar Radiation	RadAvg		1.40	2.31	3.51	4.44	5.06	5.50	5.38	4.67	3.67	2.14	1.24	0.94
	RadStd		0.19	0.28	0.28	0.38	0.53	0.35	0.53	0.30	0.31	0.18	0.15	0.09

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page